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Review

Green hydrogen as a power plant fuel: What is energy efficiency from production to utilization?

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ABSTRACT

This work is focused on analyzing the efficiency of using "green" hydrogen as a fuel for power generation systems. Three main stages of the process were considered: hydrogen production through electrolysis, hydrogen compression for transportation, and hydrogen utilization as a fuel in a combined cycle power plant. It was established that LHV efficiency of alkaline water and proton-membrane electrolysis is up to 70% and 85%, respectively. The thermodynamic analysis of hydrogen compression showed that the energy consumption per 1 kg of hydrogen to compress from 20 bar to 600 bar is 1.51 kWh with the utilization of waste-heat in an Organic Rankine Cycle (ORC), and 1.77 kWh without ORC. This energy consumption corresponds to losses of 4.5% and 5.3% of hydrogen LHV, respectively. The energy analysis of using hydrogen as a fuel was performed for pure hydrogen and hydrogen-methane blends of different compositions. An increase in the hydrogen volume fraction in the fuel increases LHV efficiency and decreases HHV efficiency. When fuel with 100% H_2 is used, HHV and LHV efficiency are 48.7% and 57.5%, respectively. The real efficiency of transforming renewable energy via hydrogen as an energy carrier into electricity in combined cycle power plants is about 38%.

1. Introduction

The energy transition is the reality in which the global energy sector currently finds itself. One of the main reasons for transitioning from fossil fuels to low-carbon energy sources is global warming and climate change [1]. There is substantial evidence supporting the causes of global warming, with greenhouse gas emissions being one of the main culprits [2,3]. Fossil fuels in the global energy and transport sectors are the main emitters of carbon dioxide. Despite advances in renewable energy development, fossil fuels (oil, natural gas, coal) still account for more than 75% of global primary energy consumption [4]. Meanwhile, the consumption of fossil fuels in absolute terms has been steadily increasing in recent years. Thus, the energy transition is a slow process.

The main reasons for the low speed of the energy transition are the relatively low cost of fossil fuels in comparison with carbon-free fuels and the long investment cycle of power equipment (for instance, the investment cycle of the steam and gas turbine power plant is more than 20 years) [5,6]. Accordingly, the power equipment for using fossil fuels will operate in the next few decades and the conventional energy sector will still play an important role.

For this reason, the question "How to reduce carbon dioxide emission in the existing energy sector, in the power plants in particular?" is raised. There are two directions for CO_2 emission reduction in the existing power plants: post-combustion [7,8] and pre-combustion [9,10]. The post-combustion way is based on the capture of CO_2 from flue gases and utilization of captured CO_2. Pre-combustion decarbonization of fuel for gas turbines is based on the concept of low- or zero-carbon fuel used for electricity generation. Two main ways of pre-combustion decarbonization of a fuel for gas turbines can be distinguished: carbon extraction and capture from a hydrocarbon fuel [11]; utilization of low-carbon (hydrogen-rich blends of carbon-containing fuels) or zero-carbon fuel (non-carbon fuel, e.g. hydrogen, ammonia, ammonia-hydrogen blends) [12–16].

The utilization of non-carbon fuels is one of the obvious ways to reduce CO_2 emissions. Examples of non-carbon fuels are including hydrogen, ammonia, ammonia-hydrogen blends [14,17,18]. Partially or fully replacement of fossil fuel with hydrogen is a promising way to decarbonize the existing power plants. The issue of the utilization of hydrogen as a fuel in the existing power plants is challenging. There

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Table 1
A color-based classification of the hydrogen production ways [23].

Color of H_2	Process	Chemical reaction	Remark
Black (Gray)	Natural gas reforming	$CH_4 + H_2O \Leftrightarrow CO + 3H_2$ $CH_4 + CO_2 \Leftrightarrow 2CO + 2H_2$	No CO_2 capture
Blue	Natural gas reforming	$CH_4 + H_2O \Leftrightarrow CO + 3H_2$ $CH_4 + CO_2 \Leftrightarrow 2CO + 2H_2$	With CO_2 capture and storage [24,25]
Green	Water electrolysis	$2H_2O \Leftrightarrow 2H_2 + O_2$	Renewable power [26,27]
Red (pink)	Water electrolysis	$2H_2O \Leftrightarrow 2H_2 + O_2$	Nuclear power [28,29]
Turquoise	Methane pyrolysis	$CH_4 \Leftrightarrow C + 2H_2$	Use of carbon as a by-product [30,31]
White	Biomass gasification	$C_xH_yO_z + O_2 \Leftrightarrow 0.5yH_2 + xCO_2$	100% of biomass as feed stock [32,33]

are attempts to use hydrogen in the existing high-power gas turbine units. In April 2022, Long Ridge Energy Terminal and General Electric reported the results of successful utilization of methane-hydrogen mixture in the combined cycle power plant [19]. The tests were performed to show the capability of the use of natural gas diluted with hydrogen. During the tests on GE7HA.02 gas turbine, the engineers used a mixture of hydrogen and methane in different proportions. Hydrogen-rich gas with 5%–25% of pure hydrogen was used as a fuel. Other sets of tests were performed to provide detailed information about the possibility of the use the hydrogen-rich fuel with 25%–50% of hydrogen.

The performances of a combined cycle power plant in Long Ridge Energy Terminal showed that hydrogen-rich gas can be considered a more climate-friendly fuel for power generation in comparison with traditional hydrocarbon fuel, e.g. natural gas. These tests were one of the first steps to getting to low-carbon energy. According to General Electric's report [19], hydrogen for dilution was produced on-site before utilization, and the volume fraction of hydrogen plans to be increased up to 100%.

There is a set of papers on the different aspects of the utilization of hydrogen as a fuel. Chiesa et al. considered the possibility of the use of hydrogen as a fuel in gas turbines [20]. The authors paid attention to the aspects of switching from natural gas to hydrogen. They considered various ways to transit from natural gas to hydrogen in the gas turbines. The results showed that the use of pure hydrogen as fuel leads to an increase in NO_x emissions. Therefore, the authors considered a gas fuel where hydrogen is diluted with steam and nitrogen. Switching from natural gas to hydrogen leads to a lower mass flow rate of the working fluid. Moreover, the composition of the working fluid is different with a higher steam fraction. In the conclusion, the authors reported that 100% hydrogen as a fuel will meet engineering problems in existing gas turbines. Therefore, the authors concluded that hydrogen diluted with inert gases such as nitrogen and steam can be used as a fuel in gas turbines.

In 2007, the U.S. Department of Energy (DOE) developed Advanced Hydrogen Turbine Development Program [21]. The program was focused to develop a combined power plant based on SGT6-6000G gas turbine fueled with hydrogen. The engineers from Siemens showed that to reduce NO_x emission, pure hydrogen should be diluted with other gases. They considered synthesis gas as a potential hydrogen-rich fuel for gas turbines to get an appropriate level of NO_x emission.

Oberg et al. recently performed a techno-economic study of gas turbines fueled with hydrogen for the low-carbon dioxide emissions systems for electricity generation [22]. The authors analyzed the blends of hydrogen and biomethane as fuel for gas turbines. They showed that utilization of hydrogen as a gas turbine fuel can be efficient with a combination of renewable methane (biomethane) even if a hydrogen cost is up to 5 €/kg_{H_2}.

Although the use of hydrogen as a fuel is promising, there are a significant number of challenges when H_2 is used. One of them is a way of hydrogen production. Currently, more than 90% of hydrogen is produced from fossil fuels. The ways of hydrogen production are classified by color differentiation. Table 1 presents the classification of the different ways of hydrogen production.

Obviously, green hydrogen is one of the most environmentally friendly fuels because carbon is omitted in the fuel cycle. However, to produce green hydrogen, renewable energy should be consumed. The main techniques of electrolysis can be highlighted: alkaline water electrolysis (AWE) [34–36], alkaline anion exchange membranes (AAEM) [37–39], solid oxide electrolysis (SOE) [40,41], and proton exchange membrane (PEM) [42,43]. The first one is capable of large-scale green hydrogen production but with moderate low efficiency. The second technique is characterized by high efficiency but with a high cost of equipment. Therefore, PEM electrolysis is acceptable for small-scale green hydrogen production.

The use of green hydrogen as a fuel for the power plant is meeting the following challenges:

- green hydrogen production via electrolysis;
- hydrogen storage and transportation;
- utilization of hydrogen as a fuel in the power plants.

Each of these challenges is characterized by low energy efficiency, which is significantly below unity. Therefore, the main goal of this paper is an assessment of the efficiency of the whole chain from green hydrogen production to green hydrogen utilization as a fuel in power plants. For this goal, the first law analysis of hydrogen production, compression, and utilization as a fuel is performed. The losses in each stage are analyzed. To analyze three main step on the way from hydrogen production to hydrogen utilization as a fuel, the thermodynamic analysis was combined with literature review. The thermodynamic analysis is applied to determine the (energy consumption of the hydrogen compression (per mass of hydrogen) and the efficiency using hydrogen is a combined power plant. The efficiency of hydrogen production was analyzed based on the literature review. Moreover, the literature review is applied to show the energy consumption of various ways of hydrogen compression.

2. Green hydrogen production: a review

Currently, over 90%–95% of hydrogen production is generated via fossil fuel reforming [23,44,45], e.g. steam methane reforming, coal gasification, etc. Only approximately 4% of hydrogen is generated via electrolysis [46]. Fig. 1 shows the global hydrogen production via various chemical techniques. It is clear that hydrogen production via electrolysis has a great potential.

Green hydrogen production via electrolysis is a process that involves splitting water molecules into hydrogen and oxygen gases using an electric current generated via renewable sources [33,48]. This technique is becoming increasingly popular as a means of producing clean and sustainable hydrogen fuel for various applications.

One example of an electrolysis technique is alkaline electrolysis, which uses a basic solution of potassium hydroxide as the electrolyte. Another example is proton exchange membrane (PEM) electrolysis, which uses a solid polymer membrane as the electrolyte.

The efficiency of electrolysis techniques can vary depending on the specific process used. Alkaline electrolysis typically has an efficiency of around 70%–80%, while PEM electrolysis can achieve efficiencies of up to 90%. Moreover, the efficiency of green hydrogen production depends on operating conditions and scale. Therefore, the initial data for the analysis in this paper were taken from the literature.

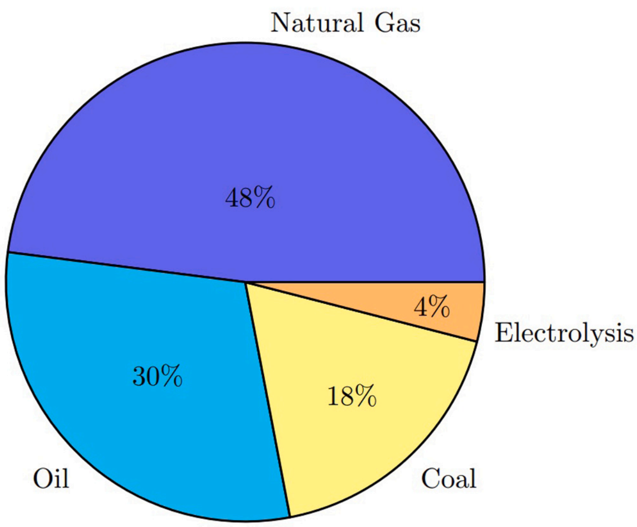

Fig. 1. Hydrogen production via various chemical techniques [47].

One advantage of hydrogen production via electrolysis is that it can be powered by renewable energy sources such as wind or solar power, making it a truly sustainable method of producing hydrogen fuel. Additionally, the only byproduct of this process is oxygen gas, which can be safely released into the atmosphere. Overall, hydrogen production via electrolysis is an important technology for enabling the widespread use of hydrogen fuel as a clean and sustainable energy source.

In the electrolysis system, the electrolyzer consists of the cells where a process of splitting water into hydrogen and oxygen is taking place by the following mechanism:

$$H_2O \longrightarrow H_2 + 0.5O_2; \tag{1}$$

$$H_2O \longrightarrow 0.5O_2 + 2H^+ + 2e^-, Anode; \tag{2}$$

$$2H^+ + 2e^- \longrightarrow H_2, Cathode. \tag{3}$$

In this paper, various techniques of green hydrogen production is reviewed. The main parameters as well as advantages and disadvantages are analyzed. The following technologies of electrolysis were analyzed:

- AWE — alkaline water electrolysis;
- AAEM — alkaline anion exchange membrane;
- PEM — proton exchange membrane;
- SOE — solid oxide electrolyzer.

2.1. AEW electrolysis

Alkaline electrolyzed water electrolysis is a process that involves splitting water molecules into hydrogen and oxygen gases using an electric current and a basic solution of potassium hydroxide as the electrolyte. The process takes place in an electrolysis cell, which consists of two electrodes – an anode and a cathode – separated by the electrolyte.

The efficiency of alkaline electrolysis can vary depending on factors such as the concentration of the electrolyte and the temperature of the system. Typically, alkaline electrolysis has an efficiency of around 70%–80%, meaning that around 20%–30% of the energy input is lost as heat.

Song et al. [49] considered the main challenges in hydrogen production from renewable energy power electrolysis. They compared alkaline water electrolysis (AWE) and proton exchange membrane (PEM) water electrolysis. the authors showed that the main advantage of alkaline electrolysis is that it is a relatively mature technology that has been used for decades in industrial applications. However, the efficiency of the existing electrolyzers is lower than the efficiency of PEM.

Yu et al. [50] provided a review on hydrogen production via alkaline water splitting and analyzed the recent progress and future perspectives of this technique. They reported that the main efforts for green hydrogen production and utilization as a fuel should be done in catalyst development and technologies of the efficient use of hydrogen. The authors showed that the voltage efficiency of AWE is 60%–80%.

Chatenet et al. [51] published a review on industrial developments of water electrolysis. They reported that metal oxides/hydroxides (especially Ni-Fe and Co-Fe oxides/hydroxides) and Ni-based alloys (especially Ni-Mo alloy) can be considered as the most perspective catalysts because these materials have shown the most activity. Moreover, these materials can be produced under mild conditions and, therefore, can be considered as a very perspective catalyst for industrial-scale production. In addition, Li et al. [52] showed that the Ni-Co-Fe-P system has the big perspectives for industrial-scale green hydrogen production.

Despite the advantages of AWE electrolysis, there are disadvantages which make the use of this technology is no widespread. The main of them is high capex, slow kinetics, high electrolyte corrosion, gas permeation. Based on the report from IRENA (International Renewable Energy Agency), the real capex in 2017 was about €750 per kW for AWE [47]. Zeng and Zhang [53] showed that slow kinetics is observed for AWE electrolysis comparing other main techniques. Liu et al. [54] reported that high electrolyte corrosion is another factor that makes industrial application of AWE is limited. Schalenbach et al. [55] showed that AWE has high gas permeation.

2.2. AAEM electrolysis

Alkaline anion exchange membrane electrolysis (AAEM) is a process that involves the use of an alkaline anion exchange membrane to separate the anode and cathode compartments in an electrolytic cell [56–58]. Advantages of AEM electrolysis include its ability to produce high-purity hydrogen gas, its scalability, and its potential for use with renewable energy sources, and relatively high efficiency compared to other hydrogen production methods. AAEM electrolysis also does not require the use of expensive catalysts, which can reduce costs. However, the cost of alkaline anion exchange membranes can be high, which can limit the widespread adoption of AAEM electrolysis for large-scale hydrogen production.

Recent advances in AAEM electrolysis include the development of new types of alkaline anion exchange membranes that have improved performance and durability [59–61]. These membranes have higher ion conductivity, lower resistance, and better chemical stability than previous generations of AAEMs.

Koshikawa et al. [62] reported that the conversion efficiency of AAEM electrolysis using NiFe-LDH and IrOx as the anode catalysts can be up to 75%. Moreover, they showed that this catalyst is moderate low cost and has a great potential to be applied for the large-scale hydrogen production.

Li and Baek [63] published a review on alkaline anion exchange membrane electrolyzers where showed the main advantages of this technique: higher efficiency comparing with AWE, more stable technology, low cost catalyst because no need noble metals, relatively low operation temperature.

Vincent and Bessarabov [64] analyzed the alkaline anion exchange membrane electrolysis as a low cost of hydrogen production technology. They highlighted the main advantages and disadvantages of this technology. The main disadvantages are the following: low OH^-–conductivity in polymeric membranes, degradation of membranes, high cost of membranes. They reported that the main challenges are dealing with membranes for AAEM.

Vijayakumar and Nam [65] showed the recommendations for future research on AAEM electrolysis. Future perspectives for AAEM electrolysis include the development of more efficient and cost-effective systems for large-scale hydrogen production. This could involve the use of renewable energy sources such as wind and solar power to power the electrolytic cells.

Overall, AAEM electrolysis has the potential to play a significant role in the transition to a low-carbon economy by providing a sustainable source of hydrogen gas for use in transportation, industry, and other applications.

2.3. PEM electrolysis

Proton exchange membrane (PEM) electrolysis is a process of splitting water into hydrogen and oxygen using an electrochemical cell. The cell consists of two electrodes, an anode and a cathode, separated by a proton exchange membrane. PEM electrolysis has high energy efficiency, with up to 85% conversion efficiency [66–68]. It also has a fast response time, allowing for quick adjustments to changes in input power. PEM electrolysis can operate at low temperatures and pressures, making it more flexible and safer than other electrolysis methods. The produced hydrogen is of high purity, making it suitable for use in fuel cells.

Chi and Yu [69] provided a comprehensive review on the different ways of hydrogen production via electrolysis. They analyzed the target price of 1 kg H2 and established that in 2020 the H2 production cost is <2.3 USD/kg_{H_2}. Moreover, they determined that electrolyzer cost is 0.5 USD/kg_{H_2}.

Falcao et al. [70] showed that PEM electrolysis has several advantages over traditional hydrogen production methods. It produces zero greenhouse gas emissions and can be powered by renewable energy sources. It is also more flexible and adaptable to different energy sources than other methods. PEM electrolysis is also scalable, making it suitable for both small-scale and large-scale hydrogen production.

Ayers et al. [71] reported about disadvantages of PEM electrolysis. The author reported that PEM electrolysis has some limitations, including high capital costs and the need for a reliable source of electricity. The proton exchange membrane can also be sensitive to impurities in the water, which can reduce its lifespan. The produced hydrogen also requires compression and storage, which can add to the overall cost. Tang et al. [72] also reported about on of the main disadvantage of PEM for large-scale application — high efficient membranes are produced using noble metals, e.g. platinum (Pt).

Jamil et al. [73] summarized the future perspectives of PEM electrolysis. It is expected that PEM will play a significant role in the transition to a low-carbon economy. Advances in technology are expected to reduce the cost of PEM electrolysis, making it more competitive with other hydrogen production methods. The development of new materials for the proton exchange membrane may also improve its durability and reduce maintenance costs [74–76].

2.4. SOE electrolysis

Hydrogen production via solid oxide electrolysis (SOE) is a promising way of green hydrogen production. The electrolyzer consists of two electrodes, an anode and a cathode, separated by a solid oxide electrolyte. SOE electrolysis has high energy efficiency, with up to 97% conversion efficiency [77–79]. It also has a fast response time, allowing for quick adjustments to changes in input power. SOE electrolysis can operate at high temperatures, making it more efficient than other electrolysis methods.

Wang et al. [80] showed the main advantages of SOE electrolysis. They reported that SOE electrolysis has several advantages over traditional hydrogen production methods. It produces zero greenhouse gas emissions and can be powered by renewable energy sources. It is also more efficient and adaptable to different energy sources than other methods. SOE electrolysis is also scalable, making it suitable for both small-scale and large-scale hydrogen production. Milewski et al. [81] calculated the electricity consumption per 1 kg of hydrogen. The authors found that 38.83 and 37.55 kWh/kg_{H_2} were is for H^+ and $O^=$ SOE, respectively.

Motylinski et al. [82] reported about the drawbacks of SOE. They reported that SOE electrolysis has some limitations, including high capital costs and the need for a reliable source of electricity. The solid oxide electrolyte can also be sensitive to impurities in the water, which can reduce its lifespan.

SOE electrolysis is still lab-scale technology and is expected to play a significant role in the transition to a low-carbon economy. Advances in technology are expected to reduce the cost of SOE electrolysis, making it more competitive with other hydrogen production methods. The development of new materials for the solid oxide electrolyte may also improve its durability and reduce maintenance costs [83–85].

2.5. Summary of electrolysis techniques

Based on the short review of the different electrolysis techniques, the main parameters were summarized in Table 2. In this table, the following parameters are provided: applicability, efficiency, charge carrier, temperature, electrolyte type, anode and cathode type, advantages and disadvantages.

Table 2 provides the conversion efficiency of the different electrolysis techniques. However, if hydrogen is used as a power plant fuel, it is important to determine the energy efficiency. Two types of electrolysis energy efficiency (η^e) can be considered: higher heating value (HHV) efficiency and lower heating value (LHV) efficiency. The equations are the following:

$$\eta^e_{HHV} = \frac{m_{H_2} \cdot HHV_{H_2}}{P_{el}}; \quad \eta^e_{LHV} = \frac{m_{H_2} \cdot LHV_{H_2}}{P_{el}} \tag{4}$$

where HHV and LHV are higher and lower heating value of hydrogen; m_{H_2} is mass flow rate of hydrogen from an electrolyzer; P_{el} is electricity for the electrolysis process.

Hydrogen combustion in gas turbines produces mainly steam. Therefore, it is expected that this steam can be condensed and additional energy output can be obtained. Based on data from Table 2, it can be concluded that the overall efficiency of green hydrogen production for an industrial application is corresponding to the efficiency of alkaline water electrolysis. Therefore, in further analysis, the efficiency of the electrolysis process is chosen as 70%. Based on Eq. (4), the specific electricity consumption per 1 kg of hydrogen can be determined. To produce 1 kg of green hydrogen, the consumption of electricity is about 47.6 kW h/kg_{H_2} (for LHV efficiency 70%). This value is corresponding to calculation of other authors [86,87]. In this case, the HHV efficiency is 82.7%.

3. Hydrogen compression

The modern tools for the thermodynamic analysis of various chemical-technological systems make it possible to determine the effect of operating parameters on the overall efficiency of the system. In this paper, Aspen HYSYS is used to simulate the following sub-systems:

- compression of hydrogen for storage and transportation;
- utilization of hydrogen in the combined cycle power plant.

Usually, green hydrogen is produced in the electrolyzer at a pressure of about 20 bar. To understand energy requirements for hydrogen compression, a thermodynamic analysis was performed. There are two main storage methods for hydrogen: gaseous hydrogen compression up to 600–700 bar and hydrogen liquefaction. Energy requirements for hydrogen liquefaction are quite high. The thermodynamic minimum in energy consumption to liquefy green hydrogen from 298 K and p = 1 bar is 3.31 kWh/kg_{H_2} [88]. Real consumption of energy for hydrogen

Table 2
The characteristics of main electrolysis techniques.

Parameter	AEW	AAEM	PEM	SOE
Applicability	Commercial (TRL9)	Laboratory (TRL6)	Near-term commercial (TRL7)	Laboratory (TRL6)
Efficiency, %	60–70	70–80	65–85	up to 97
Charge carrier	OH^-	OH^-	H^+	H^+
Temperature, °C	20–80	20–200	20–200	500–1000
Pressure, bar	3–200	1–15	10–200	10–60
Electrolyte type	Liquid	Polymeric	Polymeric	Ceramic
Anode type	Ni-Co-Fe oxides Perovskites, $LaCoO_3$	Ni-based	Ir-Ru oxides TiO_2, TiO, TiC (support)	Perovskites with proton-electronic conductivity
Cathode type	Ni-based alloys	Ni, Ni-Fe, $NiFe_2O_4$	Pt/C, MoS_2	Ni-based cermets
Advantages	Well studied, moderate low capex, stable technology	Higher efficiency, stable technology	Compact design, Higher efficiency, High flexibility	Highest efficiency, Low capex, low energy consumption
Disadvantages	Slow kinetic, High electrolyte corrosion, Gas permeation	Low OH^- conductivity in polymeric membranes, Higher capex	Higher capex, Acidic; Noble metals in membranes	Unstable process (cracking electrodes); Safety challenges

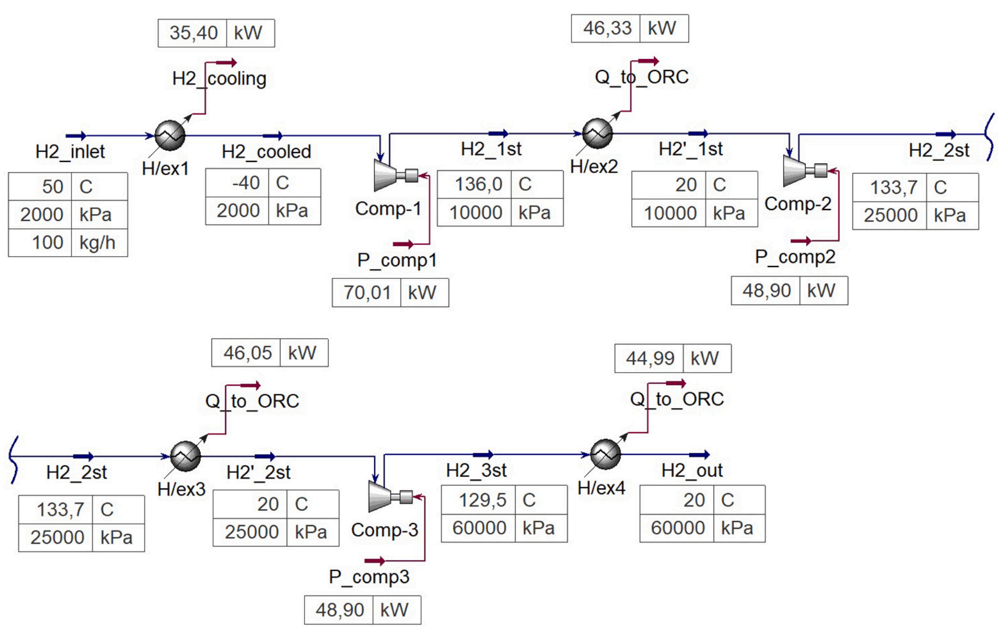

Fig. 2. The computational scheme of the hydrogen compression process: H/ex1, H/ex2, H/ex3, H/ex4 — heat exchangers; Comp-1, Comp-2, Comp-3 — compressors.

liquefaction is notably greater, typically up to 12 kWh/kg_{H_2} (depending on the process scale) [89–91]. Due to high energy requirements for hydrogen liquefaction, gaseous hydrogen compression is chosen for further analysis.

The minimum thermodynamic energy to compress hydrogen from 20 bar to 350 bar is 1.06 kWh/kg_{H_2} and from 20 bar to 600 bar is 1.27 kWh/kg_{H_2}. The actual liquefaction energy requirement is depending on a schematic diagram and the scale. In this paper, the schematic diagram with three compressors and four heat exchangers is considered. The computational scheme of the compression process is presented in Fig. 2. In this scheme, initial hydrogen at 20 bar is cooled up to −40 °C before the first compressor (Comp-1). A vapor-compression refrigeration system with a coefficient of performance of 3.5 is used for the cooling process. The temperature of hydrogen is 136, 133.7, 129.5 °C, respectively after compressors(Comp-1, Comp-2, Comp-3). After each compressor, hydrogen is cooled up to 20 °C in the heat exchangers. The cooling process is realized with organic fluid. Therefore, hot organic fluid can be utilized in Organic Rankine Cycle (ORC). The efficiency of ORC at these parameters is about 20%.

The parameters of the hydrogen compression system are summarized in Table 3.

The electricity consumption for the hydrogen compression E_{H_2} and the energy efficiency of compression (η^c) can be determined from the following equations:

Table 3
The parameters of the hydrogen compression system.

Parameter	Value
Compressor isentropic efficiency	80%
Heat exchanger efficiency	100%
COP for cooling system	3.5 °C
ORC efficiency	20%
Pressure loss	0%
Hydrogen leakage	0%

$$E_{H_2} = \sum P_{comp,i} + \frac{Q_{cool}}{COP} - \sum Q_{ORC} \cdot \eta_{ORC} \tag{5}$$

$$\eta^c_{HHV} = 1 - \frac{E_{H_2}}{m_{H_2} \cdot HHV_{H_2}}; \quad \eta^c_{LHV} = 1 - \frac{E_{H_2}}{m_{H_2} \cdot LHV_{H_2}} \tag{6}$$

Based in Eq. (5), electricity consumption to compress hydrogen from 20 bar to 600 bar is 1.51 kWh/kg_{H_2}. Electricity consumption without ORC is 1.77 kWh/kg_{H_2}. Therefore, electricity consumption is 4.5% (with ORC) and 5.3% (without ORC) of hydrogen LHV.

Fig. 3 depicts the energy consumption for hydrogen compression in three various scenarios: moderate pressure, high pressure, and liquid

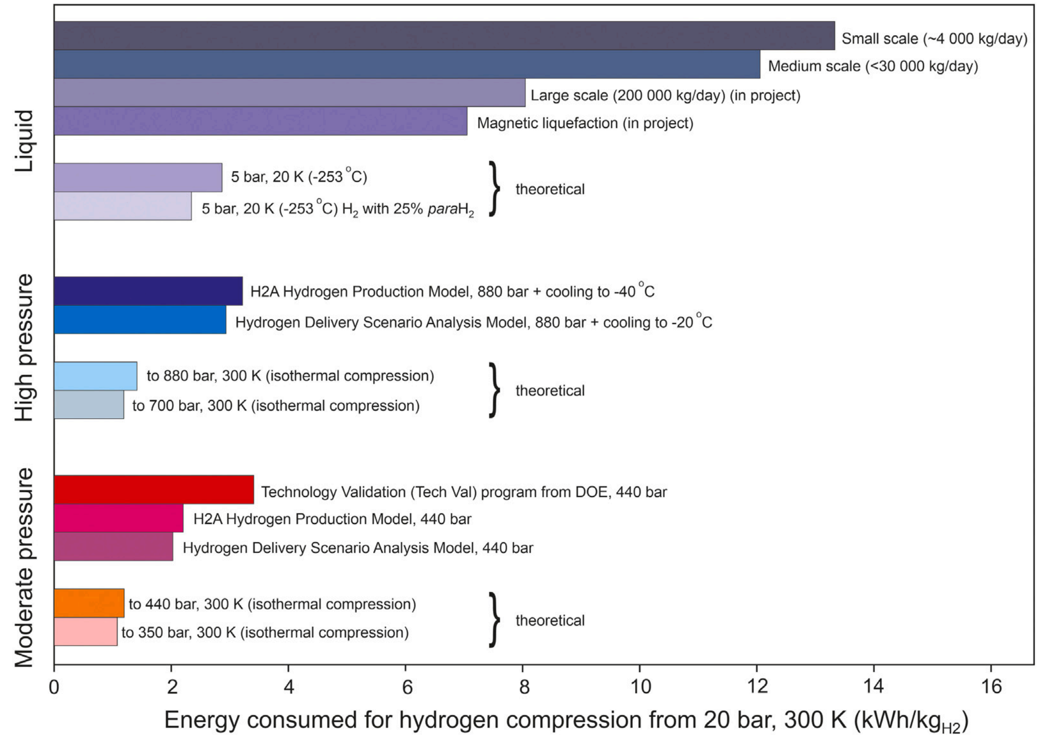

Fig. 3. Energy consumption for hydrogen compression [92–94].

hydrogen. It should be noted that the energy consumptions from Fig. 3 are the reference values. The real energy consumption is depending on various factors such as scale, climate, equipment, etc.

Based on the data from Fig. 3, we can see that H2 A Delivery Scenario Model project (U.S. Department of Energy) proposed the energy consumption of 2.30 and 3.10 kWh/kg$_{H_2}$ for hydrogen compression up to 440 bar and 880 bar. Such a high level of energy consumption is due to the low efficiency of the compressors. Compression of hydrogen occurs usually by the reciprocal compressors with a motor efficiency of 92% and an isentropic efficiency of about 55%. The energy consumption for hydrogen cooling up to −40 °C is about 0.21 kWh/kg$_{H_2}$. Hydrogen Delivery Scenario Analysis Model (HDSAM) proposed by U.S. Department of Energy has been developed a hydrogen compression station for verification with the energy consumption of 3.12 kWh/kg$_{H_2}$. Fig. 3 shows that the energy consumption varies in a wide range of data because various compressors and schematic diagrams can be used for hydrogen compression. For this reason, the development of optimal hydrogen compression schemes is an urgent task at present. Liquid hydrogen is a promising way for hydrogen storage and transportation. However, in this case, the energy consumption is more than 30% of hydrogen LHV. From an energy point of view, the liquefaction of hydrogen has extremely low energy efficiency.

3.1. Hydrogen in CCPP

One potential method of lowering CO_2 emissions is the utilization of H_2 as a fuel in power plants. Hydrogen-fueled power plants have a schematic diagram that is nearly identical to natural gas-fueled power plants. Before using hydrogen in power plants, certain systems, including the infrastructure for fuel supply, the combustion system, the heat recovery system, and the NO_x removal system, need to be updated. Natural gas and H_2 have various characteristics. As a result, the safety mechanisms of power plants that run on hydrogen may need to be modified. Moreover, the volume LHV of H_2 is almost three times lower than that of natural gas. The same of H_2 hence yields a lower value for heat.

The use of hydrogen as a fuel in the power generation plants is one of the potential ways to reduce carbon dioxide emissions. The schematic diagram of the power plants fueled with hydrogen is almost the same as for the power plants fueled with natural gas. However, some systems should be modified before hydrogen will be used in the power plants: fuel supply infrastructure, combustion system, heat recuperation system, NO_x removal system. H_2 has different properties than natural gas. Therefore, power plants fueled with hydrogen may need modification in the safety systems. Moreover, H_2 has a volume LHV that is about three times lower than that of natural gas. Thus, the same of H_2 gives a smaller value of heat. H2 molecules are also smaller than methane (CH4) molecules. Due to this reason, seal leakage is increased. Due to these difficulties, there are not any high-power reactors operating today that use fuel made entirely of H2

The use of hydrogen-rich natural gas (a mixture of natural gas and hydrogen) throughout the phase of transition to zero-carbon energy is a promising method. With the current fuel supply infrastructure, hydrogen-methane blends can be used in existing power plants. There are numerous instances of hydrogen-rich natural gas being used in gas turbines that are already in place. According to General Electric, they have roughly thirty gas turbines that run on fuels containing at least 50% hydrogen (by volume) [95].

When pure hydrogen is added to natural gas, the fuel's calorimetric characteristics change. To determine the calorimetric properties of the hydrogen-methane combination, several compositions were investigated. Fig. 4 demonstrates the carbon dioxide emission and calorimetric properties of hydrogen-rich fuel: (a) hydrogen mass fraction and carbon dioxide mass per 1 kg of H_2-rich gas as a function of hydrogen volume fraction in fuel; (b) higher heating value (HHV) and lower heating value (LHV) of 1 kg of H_2-rich gas and mass of H_2-rich gas to obtain HHV (55.5 MJ/kg) and LHV (50.0 MJ/kg) of 1 kg of methane.

The density of hydrogen is about 8 times less than the density of methane. As it can be seen from Fig. 4a, the mass fraction of hydrogen in H_2-CH_4-mixture is increasing non-linearly. 50% of H_2 volume fraction gives only 11% of H_2 mass fraction. As a consequence, carbon dioxide emission per 1 kg of hydrogen-rich fuel is also non-linear depending on H_2 volume fraction.

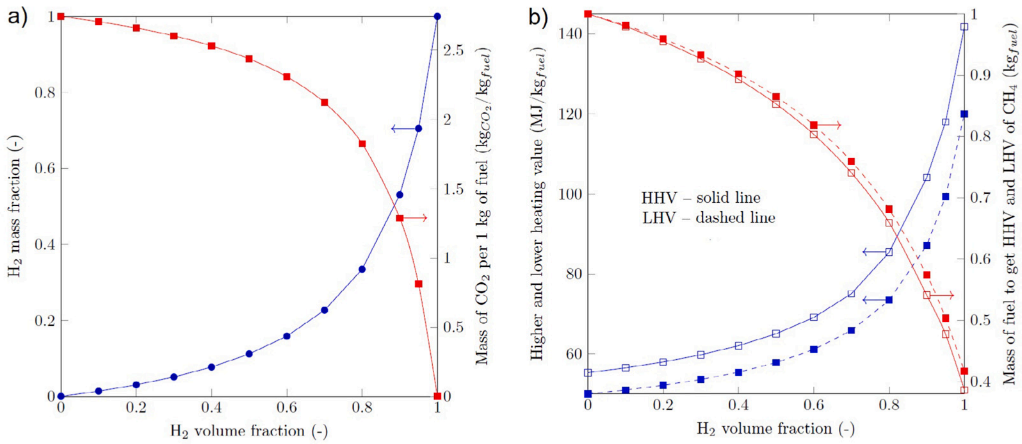

Fig. 4. HHV and LHV of hydrogen-rich gas and CO_2 emission: (a) H_2 mass fraction and mass of CO_2 per 1 kg of hydrogen-rich fuel as a function of H_2 volume fraction in fuel; (b) Higher (HHV) and lower (LHV) heating value of 1 kg of hydrogen-rich fuel and mass of hydrogen-rich fuel to obtain HHV (55.5 MJ/kg) and LHV (50.0 MJ/kg) of 1 kg of methane.

Table 4
The parameters of combines cycle power plant with the gas turbine 7HA.02.

Parameter	Value
Gas turbine power	330 MW
Gas turbine inlet temperature (TIT)	1400 °C
Compression/expansion ratio	21.5
Turbine isentropic efficiency	90%
Compressor isentropic efficiency	85%
Steam temperature at steam turbine inlet	400 °C
Steam pressure	40 bar
Condenser pressure	0.06 bar
Exhaust gas temperature after steam generator	100 °C

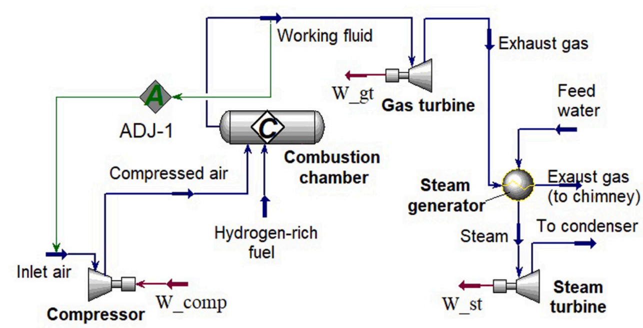

Fig. 5. A computational scheme of a combined cycle power plant fired with hydrogen-rich fuel.

About 8 times less hydrogen has a density than methane does. The mass fraction of hydrogen in the H2-CH4-mixture is rising non-linearly, as seen in Fig. 4(a). Just 11% of the H2 mass fraction is produced by 50% of the H2 volume fraction. As a result, the amount of carbon dioxide emitted per kilogram of fuel containing hydrogen depends on the volume percentage of H2.

The mass of CO2 produced by the combustion of 1 kg of hydrogen-rich fuel containing 50% (vol.) H2 is 11% less than the amount produced by the combustion of pure methane. The heat of combustion of hydrogen per kg, however, is more than twice as high as that of methane. In light of this, the mass of CO_2 per 1 MJ of the combustion heat is 42 g_{CO_2} for hydrogen-rich fuel with 50% (vol.) of H_2, compared to 54_{CO_2} for pure methane. In this case, a decrease in CO_2 emission is about 22%. The results from Fig. 4a illustrates that an increase in H_2 volume fraction above 50% will lead to a significant decrease in CO_2 emission. At the same time, an increase in H_2 volume fraction up to 50% does not play an important role in decreasing CO_2 emission.

The combined cycle power plant is built on the General Electric 7HA.02 gas turbine, which has a power output of 330 MW. In Long Ridge Energy Terminal, where the combustion of hydrogen-rich fuel was studied, the gas turbines of the 7HA-series are placed. Once exhaust gas after gas turbine has been used to generate steam for a steam turbine cycle. The combined cycle power plant's specifications are given in Table 4.

The combined cycle power plant's computational plan, which uses fuel rich in hydrogen, is depicted in Fig. 5. The turbine inlet temperature (1400 °C) was established as a constant in this design for all situations that were tested (ranging from 0 to 100% hydrogen volume fraction). The gas turbine's output was also fixed at 330 MW in all circumstances. The temperature of exhaust gas following a steam generator was set to 100 °C for all cases.

The mass flow rate of a hydrogen-rich fuel and the working fluid in gas turbines both alter when methane is completely or partially replaced by hydrogen. The performance of a combined cycle power plant and the effect of H_2 volume fraction on a mass flow rate of H_2-rich are shown in Fig. 6

Fig. 6a shows that an increase in H_2 volume fraction in a hydrogen-rich fuel leads to a decrease in a mass flow rate of fuel: 56400 kg/h for H_2 = 0%; 48400 kg/h for H_2 = 50% (16.8% less then for H_2 = 0%); 22920 kg/h for H_2 = 100% (60.4% less that for H_2 = 0%). The mass flow rate of a fuel rich in hydrogen is shown to be drastically lowering in Fig. 6a. Because the combustion heat per unit volume of H2 is roughly three times lower than the combustion heat per unit volume of CH4, the volume flow rate is rising. Fig. 6a demonstrates that the current fuel supply infrastructure of the gas turbines may be used at a low H2 volume proportion. Nonetheless, the infrastructure for fuel supply should be established for gasoline with a high H2 volume proportion.

A combined cycle power plant's performance changes when methane is diluted by hydrogen. The efficiency of a combined cycle power plant is affected by the volume proportion of H2 in a hydrogen-rich fuel, as illustrated in Fig. 6b. The following formulas can be used to calculate the fuel efficiency for lower and higher heating values [96]:

$$\eta^f_{HHV} = \frac{N_{CCPP}}{\left(\dot{m}_{H_2} \cdot HHV_{H_2} + \dot{m}_{CH_4} \cdot HHV_{CH_4}\right)};$$
$$\eta^f_{LHV} = \frac{N_{CCPP}}{\left(\dot{m}_{H_2} \cdot LHV_{H_2} + \dot{m}_{CH_4} \cdot LHV_{CH_4}\right)}; \quad (7)$$

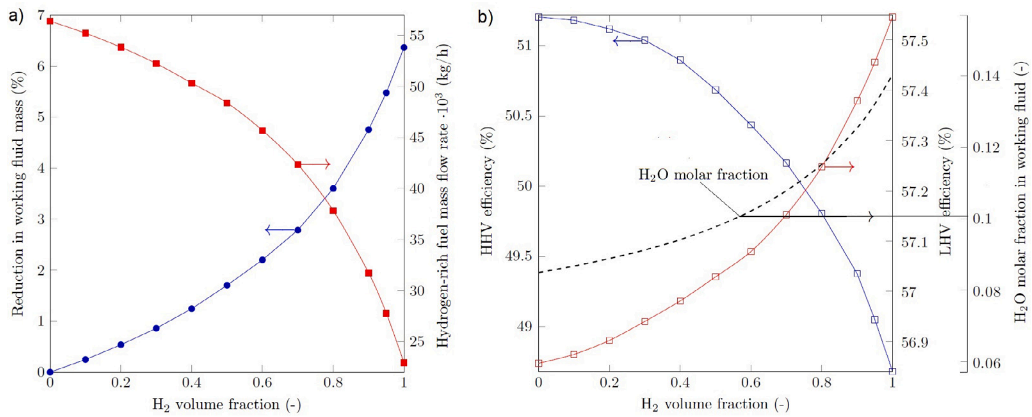

Fig. 6. H_2-rich fuel mass flow rate and reduction in a working fluid mass flow rate as a function of H_2 volume fraction (a) and the effect of H_2 volume fraction on HHV and LHV efficiency of a combined cycle power plant as well as on H_2O molar fraction in a working fluid (b).

where N_{CCPP} is power output of combined cycle power plant; $\dot{m}_{CH_4}, \dot{m}_{H_2}$ is a mass of methane and hydrogen in a hydrogen-rich fuel; LHV_i, HHV_i is lower and higher heating value of ith gas component.

Fig. 6b shows that an increase in H_2 volume fraction leads to a slight increase in fuel LHV efficiency. The difference between fuel LHV efficiency at 0 and 100% of H_2 is approximately 0.7%. However, an increase in H_2 volume fraction leads to a notable decrease in HHV efficiency (the difference is more than 2.5%). This observation is primarily the result of a working fluid's composition changing. In contrast to CH4 combustion, which creates a mixture of H_2O and CO_2, H_2O is the only product of H_2 combustion. In a working fluid, a rise in H_2 volume fraction causes an increase in H_2O molar fraction. As a result, latent heat of condensation losses are rising. The installation of the condenser heat exchangers may have been necessary due to a change in the composition of the working fluid. In this instance, the use of hydrogen-rich fuel will have an additional beneficial impact on the combined cycle power plants' overall efficiency.

4. Overall efficiency

In the above section, the efficiency of hydrogen utilization has been considered. Three main stages of using hydrogen as a fuel have been analyzed: hydrogen production, hydrogen compression, and hydrogen utilization as a fuel in CCPP.

The overall efficiency of the system "hydrogen production – hydrogen compression – hydrogen utilization as fuel" can be determined based on the following equations:

$$\eta^{total}_{HHV} = \eta^{e}_{HHV} \cdot \eta^{c}_{HHV} \cdot \eta^{f}_{HHV}; \qquad \eta^{total}_{LHV} = \eta^{e}_{LHV} \cdot \eta^{c}_{LHV} \cdot \eta^{f}_{LHV}, \tag{8}$$

where η^e, η^c, η^f are the efficiency of hydrogen production, hydrogen compression, hydrogen utilization as a fuel, respectively.

Assuming the results obtained above for the case when a fuel with 100% hydrogen is used in the combined cycle power plant, the overall efficiency can be calculated:

$$\eta^{total}_{HHV} = 0.827 \cdot 0.955 \cdot 0.487 = 0.385; \ \eta^{total}_{LHV} = 0.700 \cdot 0.949 \cdot 0.575 = 0.382 \tag{9}$$

The calculations showed that the actual efficiency of transforming renewable energy into electricity using hydrogen as an energy carrier in the combined cycle power plants is significantly less than 0.5. The efficiency of CCPP has almost reached its maximum limit, as further increasing the turbine inlet temperature is limited by turbine blade strength and NO_x formation during combustion. The practical maximum turbine inlet temperature of around 1600 °C is observed in a gas turbine produced by Mitsubishi [97]. Therefore, achieving a significant increase in the efficiency of CCPP is a difficult task. Based on Eq. (8), it can be seen that the maximum and realizable potential for increasing the overall efficiency of transforming renewable energy into electricity using hydrogen as an energy carrier in the combined cycle power plants is in the first stage — hydrogen generation via electrolysis. Even though the hydrogen compression process is not very efficient, its contribution to the overall efficiency is minimal.

5. Conclusion

This paper considered the efficiency of transformation of renewable energy into electricity when hydrogen is used as an energy carrier: electrolysis – hydrogen compression and transportation – hydrogen utilization as a fuel in a combined cycle power plant. For this goal, each stage has been thermodynamically analyzed. The LHV efficiency of hydrogen production via alkaline water electrolysis is about 70%. The LHV efficiency of the proton-membrane systems can be achieved up to 85%. The HHV efficiency of hydrogen production via electrolysis is 12.7% higher than the LHV efficiency because the combustion products of hydrogen are containing mostly steam (H_2O). Thermodynamic analysis of hydrogen compression showed that the energy consumption per 1 kg of hydrogen to compress from 20 bar to 600 bar is 1.51 kWh with the utilization of waste-heat in an Organic Rankine Cycle (ORC) and 1.77 kWh without ORC. This energy consumption is corresponding to losses of 4.5% and 5.3% of hydrogen LHV, respectively. The utilization of hydrogen as a fuel in the combined cycle power plants is a quite challenging task. Therefore, to use hydrogen as a fuel, the use of hydrogen-methane mixture as a fuel was considered. Hydrogen addition to methane leads to an insignificant effect on LHV efficiency; however, an increase in H_2 volume fraction leads to a decrease in HHV efficiency due to an increase in a volume fraction of H_2O in a working fluid. HHV and LHV efficiency of CCPP when pure H_2 is used as a fuel is 48.7% and 57.5%, respectively. Based on the analysis it was established that the real efficiency of transformation of renewable energy via hydrogen as an energy carrier into electricity in the combined cycle power plants is about 38%. Also, the directions for further improving the efficiency of such systems are described and argued.

Nomenclature

m	mass flow rate;
η	efficiency;
Pel	electric power;
E_{H_2}	electricity consumption for H_2 production;
Q	heat flow;
N	power output;
COP	coefficient of performance;
LHV	lower heating value;
HHV	higher heating value;
ORC	organic Rankine cycle;

AWE	alkaline water electrolysis;
AAEM	alkaline anion exchange membranes;
SOE	solid oxide electrolysis;
PEM	proton exchange membrane;
CCPP	combined cycle power plant.

CRediT authorship contribution statement

Dmitry Pashchenko: Conceptualization, Data curation, Formal analysis, Funding acquisition, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

[1] Jens Marquardt, Naghmeh Nasiritousi, Imaginary lock-ins in climate change politics: the challenge to envision a fossil-free future, Environ. Politics 31 (4) (2022) 621–642.

[2] Kelvin O. Yoro, Michael O. Daramola, CO2 emission sources, greenhouse gases, and the global warming effect, in: Advances in Carbon Capture, Elsevier, 2020, pp. 3–28.

[3] Loiy Al-Ghussain, Global warming: Review on driving forces and mitigation, Environ. Prog. Sustain. Energy 38 (1) (2019) 13–21.

[4] Tadeusz Olkuski, Wojciech Suwała, Artur Wyrwa, Janusz Zyśk, Barbara Tora, Primary energy consumption in selected EU countries compared to global trends, Open Chem. 19 (1) (2021) 503–510.

[5] Nazim Z. Muradov, T. Nejat Veziroğlu, Carbon-Neutral Fuels and Energy Carriers, CRC Press, 2011.

[6] Wai Siong Chai, Yulei Bao, Pengfei Jin, Guang Tang, Lei Zhou, A review on ammonia, ammonia-hydrogen and ammonia-methane fuels, Renew. Sustain. Energy Rev. 147 (2021) 111254.

[7] Mohammad Younas, Mashallah Rezakazemi, Muhammad Daud, Muhammad B. Wazir, Shakil Ahmad, Nehar Ullah, Seeram Ramakrishna, et al., Recent progress and remaining challenges in post-combustion CO2 capture using metal-organic frameworks (MOFs), Prog. Energy Combust. Sci. 80 (2020) 100849.

[8] Shihan Zhang, Yao Shen, Lidong Wang, Jianmeng Chen, Yongqi Lu, Phase change solvents for post-combustion CO2 capture: Principle, advances, and challenges, Appl. Energy 239 (2019) 876–897.

[9] Sai Gokul Subraveti, Kasturi Nagesh Pai, Ashwin Kumar Rajagopalan, Nicholas Stiles Wilkins, Arvind Rajendran, Ambalavan Jayaraman, Gokhan Alptekin, Cycle design and optimization of pressure swing adsorption cycles for pre-combustion CO2 capture, Appl. Energy 254 (2019) 113624.

[10] Carlos Arnaiz del Pozo, Schalk Cloete, Jan Hendrik Cloete, Ángel Jiménez Álvaro, Shahriar Amini, The oxygen production pre-combustion (OPPC) IGCC plant for efficient power production with CO2 capture, Energy Convers. Manage. 201 (2019) 112109.

[11] Qin Chen, Fabian Rosner, Ashok Rao, Scott Samuelsen, Ambal Jayaraman, Gokhan Alptekin, Simulation of elevated temperature solid sorbent CO2 capture for pre-combustion applications using computational fluid dynamics, Appl. Energy 237 (2019) 314–325.

[12] Pavlos Dimitriou, Rahat Javaid, A review of ammonia as a compression ignition engine fuel, Int. J. Hydrogen Energy 45 (11) (2020) 7098–7118.

[13] Krishna Kumar Gupta, Ameenur Rehman, R.M. Sarviya, Bio-fuels for the gas turbine: A review, Renew. Sustain. Energy Rev. 14 (9) (2010) 2946–2955.

[14] Dmitry Pashchenko, Ravil Mustafin, Ammonia decomposition in the thermochemical waste-heat recuperation systems: A view from low and high heating value, Energy Convers. Manage. 251 (2022) 114959.

[15] Dmitry Pashchenko, Ravil Mustafin, Igor Karpilov, Efficiency of chemically recuperated gas turbine fired with methane: Effect of operating parameters, Appl. Therm. Eng. 212 (2022) 118578.

[16] Andrey Rogalev, Evgeniy Grigoriev, Vladimir Kindra, Nikolay Rogalev, Thermodynamic optimization and equipment development for a high efficient fossil fuel power plant with zero emissions, J. Clean. Prod. 236 (2019) 117592.

[17] Dmitry Pashchenko, Hydrogen-rich gas as a fuel for the gas turbines: A pathway to lower CO2 emission, Renew. Sustain. Energy Rev. 173 (2023) 113117.

[18] Dmitry Pashchenko, Low-grade heat utilization in the methanol-fired gas turbines through a thermochemical fuel transformation, Therm. Sci. Eng. Prog. 36 (2022) 101537.

[19] Sonal Patel, First hydrogen burn at long ridge HA-class gas turbine marks triumph for GE, Powermag (2022) https://www.powermag.com/first-hydrogen-burn-at-long-ridge-ha-class-gas-turbine-marks-triumph-for-ge/.

[20] Paolo Chiesa, Giovanni Lozza, Luigi Mazzocchi, Using hydrogen as gas turbine fuel, J. Eng. Gas Turbines Power 127 (1) (2005) 73–80.

[21] Ed Bancalari, Pedy Chan, Ihor S. Diakunchak, Advanced hydrogen gas turbine development program, in: Turbo Expo: Power for Land, Sea, and Air, Vol. 47918, 2007, pp. 977–987.

[22] Simon Öberg, Mikael Odenberger, Filip Johnsson, The value of flexible fuel mixing in hydrogen-fueled gas turbines–A techno-economic study, Int. J. Hydrogen Energy (2022) http://dx.doi.org/10.1016/j.ijhydene.2022.07.075.

[23] Furat Dawood, Martin Anda, G.M. Shafiullah, Hydrogen production for energy: An overview, Int. J. Hydrogen Energy 45 (7) (2020) 3847–3869.

[24] Roghayeh Habibi, Fathollah Pourfayaz, Mehdi Mehrpooya, Hamed Kamali, A natural gas-based eco-friendly polygeneration system including gas turbine, sorption-enhanced steam methane reforming, absorption chiller and flue gas CO2 capture unit, Sustain. Energy Technol. Assess. 52 (2022) 101984.

[25] Zaira Navas-Anguita, Diego García-Gusano, Javier Dufour, Diego Iribarren, Revisiting the role of steam methane reforming with CO2 capture and storage for long-term hydrogen production, Sci. Total Environ. 771 (2021) 145432.

[26] N.A. Burton, R.V. Padilla, A. Rose, H. Habibullah, Increasing the efficiency of hydrogen production from solar powered water electrolysis, Renew. Sustain. Energy Rev. 135 (2021) 110255.

[27] F. Gutiérrez-Martín, Lidia Amodio, Maurizio Pagano, Hydrogen production by water electrolysis and off-grid solar PV, Int. J. Hydrogen Energy 46 (57) (2021) 29038–29048.

[28] Roxanne Pinsky, Piyush Sabharwall, Jeremy Hartvigsen, James O'Brien, Comparative review of hydrogen production technologies for nuclear hybrid energy systems, Prog. Nucl. Energy 123 (2020) 103317.

[29] Ali Erdogan Karaca, Ibrahim Dincer, Junjie Gu, Life cycle assessment study on nuclear based sustainable hydrogen production options, Int. J. Hydrogen Energy 45 (41) (2020) 22148–22159.

[30] Stefan Schneider, Siegfried Bajohr, Frank Graf, Thomas Kolb, State of the art of hydrogen production via pyrolysis of natural gas, ChemBioEng Rev. 7 (5) (2020) 150–158.

[31] Jarrett Riley, Chris Atallah, Ranjani Siriwardane, Robert Stevens, Technoeconomic analysis for hydrogen and carbon co-production via catalytic pyrolysis of methane, Int. J. Hydrogen Energy 46 (39) (2021) 20338–20358.

[32] Leichang Cao, K.M. Iris, Xinni Xiong, Daniel C.W. Tsang, Shicheng Zhang, James H. Clark, Changwei Hu, Yun Hau Ng, Jin Shang, Yong Sik Ok, Biorenewable hydrogen production through biomass gasification: A review and future prospects, Environ. Res. 186 (2020) 109547.

[33] H. Ishaq, I. Dincer, Comparative assessment of renewable energy-based hydrogen production methods, Renew. Sustain. Energy Rev. 135 (2021) 110192.

[34] Diogo M.F. Santos, César A.C. Sequeira, José L. Figueiredo, Hydrogen production by alkaline water electrolysis, Quím. Nova 36 (2013) 1176–1193.

[35] Jörn Brauns, Thomas Turek, Alkaline water electrolysis powered by renewable energy: A review, Processes 8 (2) (2020) 248.

[36] Stefania Marini, Paolo Salvi, Paolo Nelli, Rachele Pesenti, Marco Villa, Mario Berrettoni, Giovanni Zangari, Yohannes Kiros, Advanced alkaline water electrolysis, Electrochim. Acta 82 (2012) 384–391.

[37] R.E. Coppola, D. Herranz, R. Escudero-Cid, N. Ming, Norma Beatriz D'Accorso, P. Ocón, Graciela Carmen Abuin, Polybenzimidazole-crosslinked-poly (vinyl benzyl chloride) as anion exchange membrane for alkaline electrolyzers, Renew. Energy 157 (2020) 71–82.

[38] E. López-Fernández, C. Gómez-Sacedón, J. Gil-Rostra, J.P. Espinós, J. Javier Brey, AR González-Elipe, A. de Lucas-Consuegra, F. Yubero, Optimization of anion exchange membrane water electrolyzers using ionomer-free electrodes, Renew. Energy 197 (2022) 1183–1191.

[39] Sander Ratso, Andrea Zitolo, Maike Käärik, Maido Merisalu, Arvo Kikas, Vambola Kisand, Mihkel Rähn, Päärn Paiste, Jaan Leis, Väino Sammelselg, et al., Non-precious metal cathodes for anion exchange membrane fuel cells from ball-milled iron and nitrogen doped carbide-derived carbons, Renew. Energy 167 (2021) 800–810.

[40] Gonzalo Puig-Samper, Eleonora Bargiacchi, Diego Iribarren, Javier Dufour, Assessing the prospective environmental performance of hydrogen from high-temperature electrolysis coupled with concentrated solar power, Renew. Energy 196 (2022) 1258–1268.

[41] Sune Dalgaard Ebbesen, Mogens Mogensen, Electrolysis of carbon dioxide in solid oxide electrolysis cells, J. Power Sources 193 (1) (2009) 349–358.

[42] Weiguang Yang, Hui Guo, Fuquan Niu, Bingjie Wang, Bin Huang, Sirui Niu, Jianli Liu, Shuting Yang, Yange Yang, A novel strategy for accelerating degradation of proton exchange membranes in fuel cell, Renew. Energy (2023).

[43] Lixin Fan, Yang Liu, Xiaobing Luo, Zhengkai Tu, Siew Hwa Chan, Comparison and evaluation of mega watts proton exchange membrane fuel cell combined heat and power system under different waste heat recovery methods, Renew. Energy 210 (2023) 295–305.

[44] Jamie D. Holladay, Jianli Hu, David L. King, Yong Wang, An overview of hydrogen production technologies, Catal. Today 139 (4) (2009) 244–260.
[45] Canan Acar, Ibrahim Dincer, Review and evaluation of hydrogen production options for better environment, J. Clean. Prod. 218 (2019) 835–849.
[46] Kalathur S.V. Santhanam, Roman J. Press, Massoud J. Miri, Alla V. Bailey, Gerald A. Takacs, Introduction to Hydrogen Technology, John Wiley & Sons, 2017.
[47] Emanuele Taibi, Raul Miranda, Wouter Vanhoudt, Thomas Winkel, Jean-Christophe Lanoix, Frederic Barth, Hydrogen from Renewable Power: Technology Outlook for the Energy Transition, Hydrogen Knowledge Centre, 2018.
[48] Hirokazu Kojima, Kensaku Nagasawa, Naoto Todoroki, Yoshikazu Ito, Toshiaki Matsui, Ryo Nakajima, Influence of renewable energy power fluctuations on water electrolysis for green hydrogen production, Int. J. Hydrog. Energy 48 (12) (2023) 4572–4593.
[49] Xiaoyun Song, Danxi Liang, Jie Song, Guizhi Xu, Zhanfeng Deng, Meng Niu, Problems and technology development trends of hydrogen production from renewable energy power electrolysis-a review, in: 2021 IEEE 5th Conference on Energy Internet and Energy System Integration, EI2, IEEE, 2021, pp. 3879–3882.
[50] Zi-You Yu, Yu Duan, Xing-Yu Feng, Xingxing Yu, Min-Rui Gao, Shu-Hong Yu, Clean and affordable hydrogen fuel from alkaline water splitting: past, recent progress, and future prospects, Adv. Mater. 33 (31) (2021) 2007100.
[51] Marian Chatenet, Bruno G. Pollet, Dario R. Dekel, Fabio Dionigi, Jonathan Deseure, Pierre Millet, Richard D. Braatz, Martin Z. Bazant, Michael Eikerling, Iain Staffell, et al., Water electrolysis: from textbook knowledge to the latest scientific strategies and industrial developments, Chem. Soc. Rev. (2022).
[52] Ang Li, Ling Zhang, Fangzheng Wang, Li Zhang, Li Li, Hongmei Chen, Zidong Wei, Rational design of porous Ni-Co-Fe ternary metal phosphides nanobricks as bifunctional electrocatalysts for efficient overall water splitting, Appl. Catal. B 310 (2022) 121353.
[53] Kai Zeng, Dongke Zhang, Recent progress in alkaline water electrolysis for hydrogen production and applications, Prog. Energy Combust. Sci. 36 (3) (2010) 307–326.
[54] Taikai Liu, Xianbin Wang, Xinge Jiang, Chunming Deng, Shaopeng Niu, Jie Mao, Wei Zeng, Min Liu, Hanlin Liao, Mechanism of corrosion and sedimentation of nickel electrodes for alkaline water electrolysis, Mater. Chem. Phys. 303 (2023) 127806.
[55] Maximilian Schalenbach, Wiebke Lueke, Detlef Stolten, Hydrogen diffusivity and electrolyte permeability of the Zirfon PERL separator for alkaline water electrolysis, J. Electrochem. Soc. 163 (14) (2016) F1480.
[56] Pandiarajan Thangavel, Miran Ha, Shanmugasundaram Kumaraguru, Abhishek Meena, Aditya Narayan Singh, Ahmad M Harzandi, Kwang S Kim, Graphene-nanoplatelets-supported NiFe-MOF: high-efficiency and ultra-stable oxygen electrodes for sustained alkaline anion exchange membrane water electrolysis, Energy Environ. Sci. 13 (10) (2020) 3447–3458.
[57] Abhishek Meena, Pandiarajan Thangavel, Arun S. Nissimagoudar, Aditya Narayan Singh, Atanu Jana, Da Sol Jeong, Hyunsik Im, Kwang S. Kim, Bifunctional oxovanadate doped cobalt carbonate for high-efficient overall water splitting in alkaline-anion-exchange-membrane water-electrolyzer, Chem. Eng. J. 430 (2022) 132623.
[58] Immanuel Vincent, Eun-Chong Lee, Hyung-Man Kim, Comprehensive impedance investigation of low-cost anion exchange membrane electrolysis for large-scale hydrogen production, Sci. Rep. 11 (1) (2021) 293.
[59] Nagappan Ramaswamy, Sanjeev Mukerjee, Alkaline anion-exchange membrane fuel cells: challenges in electrocatalysis and interfacial charge transfer, Chem. Rev. 119 (23) (2019) 11945–11979.
[60] Zhengwang Tao, Chenyi Wang, Xiaoyan Zhao, Jian Li, Michael D. Guiver, Progress in high-performance anion exchange membranes based on the design of stable cations for alkaline fuel cells, Adv. Mater. Technol. 6 (5) (2021) 2001220.
[61] Vikrant Yadav, Abhishek Rajput, Prem P. Sharma, Prafulla K. Jha, Vaibhav Kulshrestha, Polyetherimide based anion exchange membranes for alkaline fuel cell: Better ion transport properties and stability, Colloids Surf. A 588 (2020) 124348.
[62] Hiroyuki Koshikawa, Hideaki Murase, Takao Hayashi, Kosuke Nakajima, Hisanori Mashiko, Seigo Shiraishi, Yoichiro Tsuji, Single nanometer-sized NiFe-layered double hydroxides as anode catalyst in anion exchange membrane water electrolysis cell with energy conversion efficiency of 74.7% at 1.0 A cm–2, ACS Catal. 10 (3) (2020) 1886–1893.
[63] Changqing Li, Jong-Beom Baek, The promise of hydrogen production from alkaline anion exchange membrane electrolyzers, Nano Energy 87 (2021) 106162.
[64] Immanuel Vincent, Dmitri Bessarabov, Low cost hydrogen production by anion exchange membrane electrolysis: A review, Renew. Sustain. Energy Rev. 81 (2018) 1690–1704.
[65] Vijayalekshmi Vijayakumar, Sang Yong Nam, Recent advancements in applications of alkaline anion exchange membranes for polymer electrolyte fuel cells, J. Ind. Eng. Chem. 70 (2019) 70–86.
[66] Xufei Gu, Zhi Ying, Xiaoyuan Zheng, Binlin Dou, Guomin Cui, Photovoltaic-based energy system coupled with energy storage for all-day stable PEM electrolytic hydrogen production, Renew. Energy 209 (2023) 53–62.
[67] Zhiqiang Xie, Shule Yu, Xiaohan Ma, Kui Li, Lei Ding, Weitian Wang, David A. Cullen, Harry M Meyer III, Haoran Yu, Jianhua Tong, et al., MoS2 nanosheet integrated electrodes with engineered 1T-2H phases and defects for efficient hydrogen production in practical PEM electrolysis, Appl. Catal. B 313 (2022) 121458.
[68] S. Shiva Kumar, V. Himabindu, Hydrogen production by PEM water electrolysis–A review, Mater. Sci. Energy Technol. 2 (3) (2019) 442–454.
[69] Jun Chi, Hongmei Yu, Water electrolysis based on renewable energy for hydrogen production, Chin. J. Catal. 39 (3) (2018) 390–394.
[70] D.S. Falcão, A.M.F.R. Pinto, A review on PEM electrolyzer modelling: Guidelines for beginners, J. Clean. Prod. 261 (2020) 121184.
[71] Katherine Ayers, The potential of proton exchange membrane–based electrolysis technology, Curr. Opin. Electrochem. 18 (2019) 9–15.
[72] Meihua Tang, Shiming Zhang, Shengli Chen, Pt utilization in proton exchange membrane fuel cells: structure impacting factors and mechanistic insights, Chem. Soc. Rev. (2022).
[73] Asif Jamil, Sikander Rafiq, Tanveer Iqbal, Hafiza Aroosa Aslam Khan, Haris Mahmood Khan, Babar Azeem, M.Z. Mustafa, Abdulkader S. Hanbazazah, Current status and future perspectives of proton exchange membranes for hydrogen fuel cells, Chemosphere (2022) 135204.
[74] Svenja Stiber, Noriko Sata, Tobias Morawietz, S.A. Ansar, Thomas Jahnke, Jason Keonhag Lee, Aimy Bazylak, Arne Fallisch, A.S. Gago, Kaspar Andreas Friedrich, A high-performance, durable and low-cost proton exchange membrane electrolyser with stainless steel components, Energy Environ. Sci. 15 (1) (2022) 109–122.
[75] Ahmad T. Mayyas, Mark F. Ruth, Bryan S. Pivovar, Guido Bender, Keith B. Wipke, Manufacturing Cost Analysis for Proton Exchange Membrane Water Electrolyzers, Technical Report, Golden, CO (United States), National Renewable Energy Lab.(NREL), 2019.
[76] Indrajit Chakraborty, Sovik Das, B.K. Dubey, M.M. Ghangrekar, Novel low cost proton exchange membrane made from sulphonated biochar for application in microbial fuel cells, Mater. Chem. Phys. 239 (2020) 122025.
[77] Yuefeng Song, Xiaomin Zhang, Kui Xie, Guoxiong Wang, Xinhe Bao, High-temperature CO2 electrolysis in solid oxide electrolysis cells: developments, challenges, and prospects, Adv. Mater. 31 (50) (2019) 1902033.
[78] Anne Hauch, Rainer Küngas, Peter Blennow, Anders Bavnhøj Hansen, John Bøgild Hansen, Brian Vad Mathiesen, Mogens Bjerg Mogensen, Recent advances in solid oxide cell technology for electrolysis, Science 370 (6513) (2020) eaba6118.
[79] Yunfeng Tian, Nalluri Abhishek, Caichen Yang, Rui Yang, Sihyuk Choi, Bo Chi, Jian Pu, Yihan Ling, John T.S. Irvine, Guntae Kim, Progress and potential for symmetrical solid oxide electrolysis cells, Matter 5 (2) (2022) 482–514.
[80] Ligang Wang, Ming Chen, Rainer Küngas, Tzu-En Lin, Stefan Diethelm, François Maréchal, et al., Power-to-fuels via solid-oxide electrolyzer: Operating window and techno-economics, Renew. Sustain. Energy Rev. 110 (2019) 174–187.
[81] Jaroslaw Milewski, Jakub Kupecki, Arkadiusz Szczesniak, Nikolaj Uzunow, Hydrogen production in solid oxide electrolyzers coupled with nuclear reactors, Int. J. Hydrogen Energy 46 (72) (2021) 35765–35776.
[82] Konrad Motylinski, Michal Wierzbicki, Jakub Kupecki, Stanislaw Jagielski, Investigation of off-design characteristics of solid oxide electrolyser (SOE) operating in endothermic conditions, Renew. Energy 170 (2021) 277–285.
[83] Aziz Nechache, Stéphane Hody, Alternative and innovative solid oxide electrolysis cell materials: A short review, Renew. Sustain. Energy Rev. 149 (2021) 111322.
[84] Yun Zheng, Zhongwei Chen, Jiujun Zhang, Solid oxide electrolysis of H 2 O and CO 2 to produce hydrogen and low-carbon fuels, Electrochem. Energy Rev. 4 (2021) 508–517.
[85] Zhao Liu, Beibei Han, Zhiyi Lu, Wanbing Guan, Yuanyuan Li, Changjiang Song, Liang Chen, Subhash C. Singhal, Efficiency and stability of hydrogen production from seawater using solid oxide electrolysis cells, Appl. Energy 300 (2021) 117439.
[86] Vladimir M. Nikolic, Gvozden S. Tasic, Aleksandar D. Maksic, Djordje P. Saponjic, Snezana M. Miulovic, Milica P. Marceta Kaninski, Raising efficiency of hydrogen generation from alkaline water electrolysis–Energy saving, Int. J. Hydrogen Energy 35 (22) (2010) 12369–12373.
[87] Xia Yanghong, Cheng Haoran, He Hanghang, Wei Wei, Efficiency and consistency enhancement for alkaline electrolyzers driven by renewable energy sources, Commun. Eng. 2 (22) (2023) http://dx.doi.org/10.1038/s44172-023-00070-7.
[88] Walter Peschka, Liquid Hydrogen: Fuel of the Future, Springer Science & Business Media, 2012.
[89] Songwut Krasae-in, Jacob H. Stang, Petter Neksa, Development of large-scale hydrogen liquefaction processes from 1898 to 2009, Int. J. Hydrogen Energy 35 (10) (2010) 4524–4533.
[90] U. Cardella, L. Decker, J. Sundberg, H. Klein, Process optimization for large-scale hydrogen liquefaction, Int. J. Hydrogen Energy 42 (17) (2017) 12339–12354.
[91] Saif Al Ghafri, Stephanie Munro, Umberto Cardella, Thomas Funke, William Notardonato, J.P. Martin Trusler, Jacob Leachman, Roland Span, Shoji Kamiya, Garth Pearce, et al., Hydrogen liquefaction: a review of the fundamental physics, engineering practice and future opportunities, Energy Environ. Sci. (2022).

[92] Giuseppe Sdanghi, Gaël Maranzana, Alain Celzard, Vanessa Fierro, Review of the current technologies and performances of hydrogen compression for stationary and automotive applications, Renew. Sustain. Energy Rev. 102 (2019) 150–170.
[93] Monterey Gardiner, Energy requirements for hydrogen gas compression and liquefaction as related to vehicle storage needs, DOE Hydrog. Fuel Cells Program Rec. 9013 (2009).
[94] Zhao Yanxing, Gong Maoqiong, Zhou Yuan, Dong Xueqiang, Shen Jun, Thermodynamics analysis of hydrogen storage based on compressed gaseous hydrogen, liquid hydrogen and cryo-compressed hydrogen, Int. J. Hydrogen Energy 44 (31) (2019) 16833–16840.
[95] https://info.gepower.com/hydrogen-and-gas-turbines-registration-page.html?gecid=H2_int_ge-gas-power_h2homepagepromo. (Online; accessed 27 August 2022).
[96] Janusz Kotowicz, Marcin Job, Mateusz Brzkeczek, Krzysztof Nawrat, Janusz Mkedrych, The methodology of the gas turbine efficiency calculation, Arch. Thermodyn. (4) (2016).
[97] Satoshi Hada, Kazumasa Takata, Yoshifumi Iwasaki, Masanori Yuri, Junichiro Masada, High-efficiency gas turbine development applying 1600 C class" J" technology, Mitsubishi Heavy Ind. Tech. Rev. 52 (2) (2015) 2.